

Plugins

runtime extensions

- **hot-reload host** · fault isolation
- UI slot injection · per-plugin i18n
- per-plugin settings
- **ctx.services** — shared capability bus
- no direct plant I/O

Recipes

versioned set-points

- line / product / recipe / run tree
- versioned · non-destructive revert
- known-good snapshot
- window $W_r(p) \cap$ hard range
- active run governs every write

AgentTeam

governed bridge to the plant

Lead — supervision

1 s tick · fingerprint skip
task tree · 7 states

Workers × N

14 harness engines
interchangeable

Cross-channel bus

lead-gated

Fallback — rule-guaranteed

FIFO dispatch · shortest queue
longest idle · 3 retries
300 s stall watchdog

Shared memory — hybrid retrieval

FTS5 lexical + vector · RRF $k = 60$
MMR $\lambda = 0.7$ · 3-layer injection · compact 70%

Tools

48-tool surface · 23 industrial/AML

- **dcw_control** — open optimization record
- **daq_query** — bounded history read
- **dcw_judge** — judges but cannot actuate
- **recipe_update** · **rollback**
- cards: unit · range · window · freshness

3 govern

interlock · approval
readback

2

propose

bounded rollback ($K = 2$)

actuate 4

PLC Nodes

5 protocol drivers

- Modbus TCP/RTU · OPC UA · MQTT · REST
- per-node edge runtime
- intervals: acq 5 s · WS 1 s · trend 5 s
- gateway sweep: rotating cursor · quota 64
- alarms: band 8% · hyst 2% · debounce 3

6

learn

5

verify · judge

1

observe